

Mental models

- Users develop an understanding of a system through learning about and using it
- Knowledge is sometimes described as a mental model:
 - How to use the system (what to do next)
 - What to do with unfamiliar systems or unexpected situations (how the system works)
- People make inferences using mental models of how to carry out tasks

Mental models

- Craik (1943) described mental models as:
 - internal constructions of some aspect of the external world enabling predictions to be made
- Involves unconscious and conscious processes
 - images and analogies are activated
- Deep versus shallow models
 - e.g. how to drive a car and how it works

Everyday reasoning and mental models

- (a) You arrive home on a cold winter's night to a cold house. How do you get the house to warm up as quickly as possible? Set the thermostat to be at its highest or to the desired temperature?
- (b) You arrive home starving hungry. You look in the fridge and find all that is left is an uncooked pizza. You have an electric oven. Do you warm it up to 375 degrees first and then put it in (as specified by the instructions) or turn the oven up higher to try to warm it up quicker?

Heating up a room or oven that is thermostat-controlled

- Many people have erroneous mental models (Kempton, 1996)
- Why?
 - General valve theory, where ‘more is more’ principle is generalised to different settings (e.g. gas pedal, gas cooker, tap, radio volume)
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 - Thermostats based on model of on-off switch model

Heating up a room or oven that is thermostat-controlled

- Same is often true for understanding how interactive devices and computers work:
 - poor, often incomplete, easily confusable, based on inappropriate analogies and superstition (Norman, 1983)
 - e.g. elevators and pedestrian crossings - lot of people hit the button at least twice
 - Why? Think it will make the lights change faster or ensure the elevator arrives!